



Archives of Environmental Health: An International Journal

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/vzeh20>

Short-Term Associations between Emergency Hospital Admissions for Respiratory and Cardiovascular Disease and Outdoor Air Pollution in London

Richard W. Atkinson^a, Stephen A. Bremner^a, H. Ross Anderson^a, David P. Strachan^a, J. Martin Bland^a & A. Ponce de Leon^b

^a Department of Public Health Sciences, St. George's Hospital, Medical School, London

^b Departamento de Estatística, IME/UERJ Rua São Francisco Xavier Maracana, Rio de Janeiro, Brazil

Published online: 05 Apr 2010.

To cite this article: Richard W. Atkinson, Stephen A. Bremner, H. Ross Anderson, David P. Strachan, J. Martin Bland & A. Ponce de Leon (1999): Short-Term Associations between Emergency Hospital Admissions for Respiratory and Cardiovascular Disease and Outdoor Air Pollution in London, Archives of Environmental Health: An International Journal, 54:6, 398-411

To link to this article: <http://dx.doi.org/10.1080/00039899909603371>

PLEASE SCROLL DOWN FOR ARTICLE

Full terms and conditions of use: <http://www.tandfonline.com/page/terms-and-conditions>

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden.

The publisher does not give any warranty express or implied or make any representation that the contents will be complete or accurate or up to date. The accuracy of any instructions, formulae, and drug doses should be independently verified with primary sources. The publisher shall not be liable for any loss, actions, claims, proceedings, demand, or costs or damages whatsoever or howsoever caused arising directly or indirectly in connection with or arising out of the use of this material.

Short-Term Associations between Emergency Hospital Admissions for Respiratory and Cardiovascular Disease and Outdoor Air Pollution in London

RICHARD W. ATKINSON
STEPHEN A. BREMNER
H. ROSS ANDERSON
DAVID P. STRACHAN
J. MARTIN BLAND
Department of Public Health Sciences
St. George's Hospital Medical School
London

A. PONCE DE LEON
Departamento de Estatística
IME/UERJ Rua São Francisco Xavier
Maracana Rio de Janeiro, Brazil

ABSTRACT. There are concerns about the possible short-term effects of outdoor air pollution on health in the United Kingdom. In a study conducted during the time period between 1987 and 1992, investigators determined that ozone had small, but significant effects on emergency respiratory admissions. In the current study, the authors investigated associations between emergency admissions and outdoor air pollution for the time period from 1992 to 1994, inclusive, and compared the results with those obtained in the earlier study. The authors also examined particulate matter less than 10 μm in diameter (PM_{10}) and carbon monoxide in the current study. Appropriate confounding factors, such as seasonal patterns, temperature, and humidity, were controlled for, and the authors used Poisson regression to estimate the association between daily emergency admissions for respiratory and cardiovascular diseases and ozone, nitrogen dioxide, sulfur dioxide, carbon monoxide, particles measured as Black Smoke, and PM_{10} . Significant positive associations were found between emergency hospital admissions for respiratory disease and PM_{10} and sulfur dioxide, but such an association did not exist for ozone. The results were not significantly different from earlier results from London and were comparable with those determined in North America and Europe. Cardiovascular disease was associated with carbon monoxide and Black Smoke, but weaker associations existed with the other pollutants studied.

CONCERNS EXIST about the possible short-term effects of current levels of outdoor air pollutants on health in the United Kingdom. During the first half of this century, very high levels of sulfur dioxide (SO_2) and particles produced from coal burning were responsible for severe health effects.¹ The measures introduced in the Clean Air Act of 1956—and in subsequent acts—were successful in reducing emissions to levels below those thought to have adverse effects on health. However, evidence from the United States obtained via use of advanced regression techniques has suggested that small associations between both daily mortality and hospital

admissions and air pollution remained detectable.^{2–6} These observations, fueled by more-diffuse concerns about the growth of vehicle pollution, stimulated many studies worldwide,^{7–11} the results of which have generally confirmed reports from the United States. The most-studied pollutants have been particulates less than 10 μm in diameter (PM_{10}) and ozone (O_3), but nitrogen dioxide (NO_2), sulfur dioxide (SO_2), and carbon monoxide (CO) have also been investigated.^{12–15}

From an earlier time-series study linked to the Air Pollution and Health: A European Approach (APHEA) project,^{16,17} investigators have known that daily admissions

to hospitals for respiratory diseases in London (1987–1992) were associated with daily O_3 concentrations.¹⁸ This result accorded with findings from other European studies that were part of the APHEA project¹¹ and was roughly comparable with findings determined in the United States. The meta-analysis of five cities in the APHEA group¹¹ also showed a smaller, less-consistent effect of particles (i.e., measured as Black Smoke [BS] or total suspended particles [TSPs]) than reported in the United States. The only study of PM_{10} in the United Kingdom was conducted in Birmingham.¹⁹ A significant association between emergency hospital admissions for respiratory diseases and PM_{10} measured on the same day was found. In the present article, we conducted an analysis during the period 1992–1994 (i.e., 3-mo overlap with the earlier London study), and this enabled us to retest associations for a different time period. Measures of PM_{10} are available in London from 1992; therefore, we were able to compare results for PM_{10} with those for BS. In addition, we included a comprehensive range of diagnoses—including cardiovascular diseases—and age groups to afford us the opportunity to determine which groups are at greatest risk.

Method

Hospital admission data. The number of emergency admissions to hospitals in Greater London for respiratory and cardiovascular diseases were obtained from the Hospital Episode Statistics (HES) system. Daily counts of admissions for specific diagnoses and age groups were constructed for the period of 1992–1994. The diagnostic groups were as follows: all respiratory diseases (ICD9 460–519); asthma (ICD9 493); chronic obstructive pulmonary disease (COPD) plus asthma (ICD9 490–496); lower respiratory infections (ICD9 466, 480–486); all cardiovascular diseases (ICD9 390–459); and ischemic heart disease (ICD9 410–414). The age groups analyzed for each diagnosis varied, but they included “all ages,” 0–14 y, 15–64 y, 0–64 y, 65+ y, 65–74 y, and 75+ y. A total of 20 diagnostic/age groups were constructed. Only admissions of patients who were residents in London and who were treated in London hospitals were used. The London hospitals serve approximately 7 million individuals who reside throughout a roughly circular area in the Thames basin (approximately 1 600 km²), as determined by the 1991 census.

Air pollution and meteorologic data. Daily maximum and minimum temperatures and 6 A.M., and 3 P.M. humidity measures at Holborn, Central London, were obtained from the Meteorological Office. Daily average temperature and humidity measures were computed as the mean of the two respective values. Air-pollution data were obtained from several monitoring stations across London, according to availability and completeness during 1992–1994 and their geographic location. The NO_2 and CO data were obtained from 3 sites, O_3 data were obtained at 2 sites, and PM_{10} data were provided by only 1 site. Black Smoke and SO_2 measurements were obtained at 5 sites. We used a standard procedure²⁰ to estimate missing values, and we calculated

single daily average values in London for each pollutant. Additional detailed information with respect to sites and measurement methods are provided in a recent comprehensive review of U.K. air quality.²¹

Statistical methods. The statistical analysis was based on the methods developed by the APHEA project.¹⁷ We used appropriate trigonometric variables (sine and cosine terms), determined with spectral analysis, to model long-term seasonal patterns. Dummy variables accounted for “day-of-week” patterns and long-term trends in the data. We also included counts of admissions for flu to allow for possible confounding resulting from influenza. The suitability of the statistical model was assessed with the overdispersion parameter, model deviance, and plots of Pearson residuals. We used the partial autocorrelation function to assess serial correlation. Once the long-term patterns in the data were accounted for, we determined the relationships with temperature and humidity, using plots of model residuals against various measures of temperature and humidity. We compared alternative methods of modeling these meteorological variables (i.e., linear, quadratic, piecewise, spline functions²²) by using model fit statistics and the most appropriate measures were selected for inclusion in the “core” model. Any remaining serial correlation was noted for subsequent adjustment in the final Poisson model.

Once the explanatory variables of the “core” model had been decided upon, an air-pollution indicator was added. Poisson regression, which allowed for overdispersion and autocorrelation,²³ was used to determine the percentage change in the mean number of admissions associated with an increase in the pollution measure. The 10th- to 90th-percentile increase in the pollutant was used because it represents well the range of pollution levels experienced in London during the analysis period, and it facilitates meaningful comparisons with results for other pollutants. A predetermined set of daily pollution measures were tested, in turn, in the statistical model. The 24-h average was used for all pollutants except O_3 , for which the maximum 8-h average was used; maximum 1-h measures were also used for NO_2 and O_3 . These measures on the same day and 1, 2, and 3 d prior to the day of visit (termed *single-day lag 0 to lag 3*), as well as cumulative lags (calculated as the mean of lags 0–1, lags 0–2, and lags 0–3) were tested. This process of “core” model building followed by pollutant testing was repeated for each outcome and age group studied. For each pollutant we selected the most significant single day lag, irrespective of direction, for reporting purposes and for further analysis in seasonal and, where appropriate, 2-pollutant models. We investigated possible interactions between seasons and used a dummy variable to indicate the season (i.e., warm season defined as April–September and the cool season as the remaining months) and, where appropriate, we also examined 2-pollutant measures to determine whether one pollutant was more important than another.

To investigate possible interactions between temperature, humidity, and air pollution, we examined the two main series (i.e., admissions for all respiratory disease

and cardiovascular disease) in further detail to investigate whether there were significant changes in the pollutant estimates, by tertile of temperature and of humidity. We also used these series to study dose-response relationships for each of the 6 pollutants.

Given the numerous tests and the selective reporting and testing of the most-significant results, statistical significance was defined as $p < .01$. The usual 95% confidence limits, however, were retained when we presented the results of the analyses both in tabular format and graphically. All statistical analyses were conducted using SAS.²⁴

Results

During the 3 y of the study (1 096 d), there were 165 032 emergency admissions for respiratory diseases and 189 109 admissions for cardiovascular diseases.

Summary statistics for the main diagnoses and for age groups studied are provided in Table 1. For the purposes of simplifying presentation and discussion in this study, we focused upon the results for the single-day measures of pollution. Results for cumulative measures are not shown (available from authors upon request).

Summary statistics for the air-pollution and meteorological variables are also shown in Table 1. The correlation between pollutants was generally strong and positive: for NO₂, the correlations with other pollutants—excluding O₃—were between 0.6 and 0.7; for BS, they were between 0.6 and 0.9; for PM₁₀, between 0.6 and 0.7; and for SO₂, between 0.5 and 0.6. Ozone was associated negatively with all the other pollutants, but seasonal differences with weak positive associations in the warm season and weak negative associations in the cool season were noted.

Table 1.—Summary of Statistics for Hospital Admissions, by Complaint and Age Group, and for Pollution and Meteorological Variables between January 1992 and December 1994

Variable	n	$\bar{x}$	SD	Minimum	Percentile			Maximum
					10th	50th	90th	
Emergency admissions (number/day)								
All respiratory diseases								
All ages	1 096	150.6	42.9	68	102	142	210	318
0–14 y	1 096	52.6	22.5	13	29	48	83	165
15–64 y	1 096	39.4	10.3	17	27	38	53	112
65+ y	1 096	58.5	19.1	13	38	55	82	150
Asthma								
All ages	1 096	38.7	13.6	7	23	36	56	98
0–14 y	1 096	20.5	10.4	2	10	18	32	78
15–64 y	1 096	14.2	5.8	3	8	13	21	85
65+ y	1 096	3.2	2.0	0	1	3	6	16
COPD + Asthma								
65+ y	1 096	22.9	7.6	7	14	22	33	63
Lower respiratory infections								
65+ y	1 096	16.7	6.6	2	10	16	25	47
All cardiovascular diseases								
All ages	1 096	172.5	31.5	72	129	176	212	258
0–64 y	1 096	54.5	12.0	22	38	55	70	91
65+ y	1 096	117.8	22.1	49	87	120	145	189
Ischemic heart disease								
0–64 y	1 096	24.5	6	8	17	24	32	49
65+ y	1 096	37.6	7.8	14	28	37	48	70
Pollutants (daily measure)								
NO ₂ —1 hr (ppb)	1 096	50.3	17.0	22.0	34.3	47.0	70.3	224.3
O ₃ —8 hr (ppb)	1 067	17.5	11.5	1.9	4.4	16.0	30.1	79.9
SO ₂ —24 hr (μg/m ³)	1 096	21.2	7.8	7.4	13.0	19.8	31.0	82.2
CO—24 hr (ppm)	1 096	0.8	0.4	0.2	0.5	0.7	1.3	5.6
PM ₁₀ —24 hr (μg/m ³)	997	28.5	13.7	6.8	15.8	24.8	46.5	99.8
BS—24 hr (μg/m ³)	1 096	12.7	7.9	1.6	5.5	10.8	21.6	69.8
Meteorological (daily measures)								
Temperature (°C)	1 096	11.9	5.0	−0.8	5.6	11.7	18.6	25.5
Humidity (%)	1 090	70.4	11.0	33.0	56.0	70.0	85.0	97.0

Notes: SD = standard deviation, $\bar{x}$ = mean, COPD = chronic obstructive pulmonary disease, NO₂ = nitrogen dioxide, O₃ = ozone, SO₂ = sulfur dioxide, CO = carbon monoxide, PM₁₀ = particulate matter 10 μm or less in diameter, and BS = Black Smoke.

Table 2.—Summary of Single-Pollutant Results for Respiratory Complaints

Age group (y)	Pollutant	All respiratory admissions			Asthma admissions			COPD + asthma admissions			Lower respiratory admissions		
		Single lag	% change in admissions (95% CI)	p	Single lag	% change in admissions (95% CI)	p	Single lag	% change in admissions (95% CI)	p	Single lag	% change in admissions (95% CI)	p
All	NO ₂	1	1.64 (0.14, 3.15)	.03	0	1.80 (−0.77, 4.44)	.2	3	3.53 (0.64, 6.50)	.02	3	3.47 (0.08, 6.97)	.04
	O ₃	1	1.21 (−1.01, 3.48)	.3	2	−3.11 (−7.07, 1.02)	.1	0	3.82 (−1.17, 9.06)	.1	0	5.58 (0.77, 10.62)	.02
	SO ₂	1	2.01 (0.29, 3.76)	.02	1	3.38 (0.42, 6.43)	.03	3	1.53 (−1.83, 5.00)	.4	3	5.16 (1.19, 9.28)	.01
	CO	1	0.74 (−0.57, 2.07)	.3	1	1.86 (−0.51, 4.29)	.1	3	1.51 (−1.05, 4.13)	.2	0	−4.14 (−7.02, −1.17)	.007
	PM ₁₀	1	2.98 (1.06, 4.94)	.002	3	2.10 (−1.12, 5.42)	.2	3	5.21 (1.60, 8.96)	.004	3	4.64 (0.58, 8.85)	.02
0–14	BS	1	1.16 (−0.37, 2.72)	.1	1	1.72 (−1.07, 4.59)	.2	3	−1.45 (−4.47, 1.65)	.4	3	3.00 (−0.42, 6.54)	.09
	NO ₂	2	1.94 (−0.39, 4.32)	.1	3	2.11 (−1.42, 5.77)	.2						
	O ₃	0	−1.62 (−5.66, 2.60)	.4	2	−6.23 (−11.25, −0.94)	.02						
	SO ₂	1	5.14 (2.59, 7.76)	.00006	1	6.74 (2.92, 10.69)	.0004						
	CO	0	1.51 (−0.47, 3.53)	.1	3	2.29 (−0.81, 5.50)	.1						
15–64	PM ₁₀	1	4.85 (2.07, 7.72)	.0006	3	3.29 (−0.74, 7.49)	.1						
	BS	0	1.74 (−0.56, 4.10)	.1	3	2.48 (−1.06, 6.14)	.2						
	NO ₂	1	1.61 (−0.82, 4.09)	.2	1	5.08 (0.81, 9.53)	.02						
	SO ₂	3	−1.98 (−5.26, 1.40)	.2	1	2.12 (−4.84, 9.60)	.6						
	CO	3	1.90 (−0.79, 4.66)	.2	3	4.58 (−0.18, 9.57)	.06						
65+	PM ₁₀	3	1.00 (−1.05, 3.10)	.3	2	3.98 (0.23, 7.86)	.04						
	BS	2	4.24 (1.26, 7.30)	.005	3	5.71 (0.67, 10.99)	.03						
	NO ₂	3	1.05 (−1.33, 3.50)	.4	2	2.37 (−2.02, 6.95)	.3						
	O ₃	3	2.53 (0.58, 4.52)	.01	3	4.53 (−2.36, 11.91)	.2						
	SO ₂	0	2.19 (−1.56, 6.09)	.3	1	−7.41 (−15.91, 1.96)	.1						

Notes: Associations are expressed as percentage change (95% confidence interval [CI]) in number of emergency admissions for a 10th- to a 90th-percentile change in each pollutant presented at its most significant single-day lag. All p values have been rounded to one significant digit. COPD = chronic obstructive pulmonary disease, NO₂ = nitrogen dioxide, O₃ = ozone, SO₂ = sulfur dioxide, CO = carbon monoxide, PM₁₀ = particulate matter 10 µm or less in diameter, and BS = Black Smoke.

All respiratory diseases. The associations between the various air pollutants and admissions for all respiratory diseases are summarized in Table 2. For each pollutant studied, only the most significant associations from the single-day lags tested, irrespective of direction, were reported. In Figure 1, the results for the all-age group are illustrated. Also shown is the pattern of associations across the various lags tested.

Generally, respiratory admissions were associated positively with the 6 pollutants studied—significantly so for PM₁₀ and SO₂ in specific age groups. In the all-ages group, only PM₁₀ was associated significantly with emergency respiratory admissions and only for the lag 1 d measure (2.98% [95% confidence interval {CI} = 1.06, 4.94; $p = .002$] increase in admissions for a 10th- to 90th-percentile increase in PM₁₀). This association with PM₁₀, however, was found in each of the age groups investigated and was particularly strong in children (lag 1 d, 4.85%; 95% CI = 2.07, 7.72; $p = .00006$). In contrast to this, we failed to find any evidence of associations between respiratory admissions and BS. There was some suggestion in the all-ages results that SO₂ was associated significantly with increases in admissions. A highly significant association with SO₂ in children was found (lag 1 d, 5.14%; 95% CI = 2.59, 7.76; $p = .00006$). There was no evidence that this association existed in the other age subgroups. A result of particular note was the lack of association between daily respiratory admissions and O₃. We failed to find any evidence of an association with O₃ for any age group, and in children (i.e., 0–14 y of age) and adults (i.e., 15–64 y of age) the associations were negative, though not significant, in direction.

Asthma. The results for asthma are also given in Table 2. All pollutant measures, except O₃, were associated positively with admissions for asthma in the all-ages group, although none of these associations were statistically significant. The results for each single-day lag tested for the all-ages group are shown in Figure 2. The associations tended to have larger effect estimates in the age subgroups, but were mostly of marginal statistical significance. However, a highly significant association between asthma admissions and SO₂ in children was found (lag 1 d, 6.74%; 95% CI = 2.92, 10.69; $p = .0004$). Comparable effect estimates were found in other age groups, but they did not reach statistical significance. No significant associations with O₃ were found.

COPD and asthma. The associations between the various air pollutants and admissions for COPD plus asthma in the elderly (65+ y) are summarized in Table 2 and Figure 3. Only PM₁₀ was associated significantly with increases in admissions in the elderly for COPD and asthma (lag 3 d, 5.21%; 95% CI = 1.60, 8.96; $p = .004$). The associations with O₃ were consistently positive but were not statistically significant. The data in Figure 3 suggest that, for a number of pollutants, the associations with COPD and asthma may have been stronger for lags exceeding 3 d.

Lower respiratory infections. Admissions for lower respiratory infections in the elderly were associated positively with increases in all pollutants (Table 2), although these were of low statistical significance. There was a gradient in the size of the associations over the lags (Fig. 4). For all pollutants, except O₃, the magnitude and strength of the associations increased as the

Table 3.—Summary of Single-Pollutant Results for Cardiovascular Complaints

Age group (y)	Pollutant	All cardiovascular diseases			Ischemic heart disease		
		Single lag	% change in admissions (95% CI)	p	Single lag	% change in admissions (95% CI)	p
All	NO ₂	0	1.35 (0.22, 2.49)	.02			
	O ₃	2	2.34 (0.19, 4.55)	.03			
	SO ₂	0	1.57 (0.22, 2.93)	.02			
	CO	0	1.27 (0.20, 2.36)	.02			
	PM ₁₀	0	1.94 (0.56, 3.33)	.006			
	BS	0	1.89 (0.64, 3.15)	.003			
0–64	NO ₂	0	1.20 (–0.62, 3.05)	.2	0	1.53 (–1.22, 4.37)	.3
	O ₃	2	–2.59 (–4.78, –0.34)	.02	0	–2.41 (–5.62, 0.92)	.2
	SO ₂	0	2.44 (0.30, 4.63)	.02	2	–2.03 (–5.35, 1.41)	.2
	CO	3	1.61 (–0.08, 3.33)	.06	0	0.89 (–1.71, 3.57)	.5
	PM ₁₀	0	3.43 (1.25, 5.65)	.002	0	4.13 (0.78, 7.59)	.02
	BS	0	2.00 (0.03, 4.01)	.05	0	1.78 (–1.20, 4.85)	.2
65+	NO ₂	0	1.68 (0.32, 3.06)	.02	0	3.03 (0.87, 5.24)	.006
	O ₃	2	3.38 (1.12, 5.69)	.003	3	–2.24 (–5.30, 0.91)	.2
	SO ₂	0	1.72 (0.15, 3.32)	.03	0	3.10 (0.61, 5.65)	.01
	CO	0	1.67 (0.40, 2.95)	.01	3	2.27 (0.19, 4.39)	.03
	PM ₁₀	0	1.53 (–0.13, 3.22)	.07	0	3.02 (0.47, 5.64)	.02
	BS	0	2.71 (1.22, 4.23)	.0004	3	2.92 (0.55, 5.34)	.02

Notes: Associations are expressed as percentage change (95% confidence interval [CI]) in number of emergency admissions for a 10th- to a 90th-percentile change in each pollutant presented at its most significant single day lag. All p values have been rounded to one significant digit. NO₂ = nitrogen dioxide, O₃ = ozone, SO₂ = sulfur dioxide, CO = carbon monoxide, PM₁₀ = particulate matter 10 μ m or less in diameter, and BS = Black Smoke.

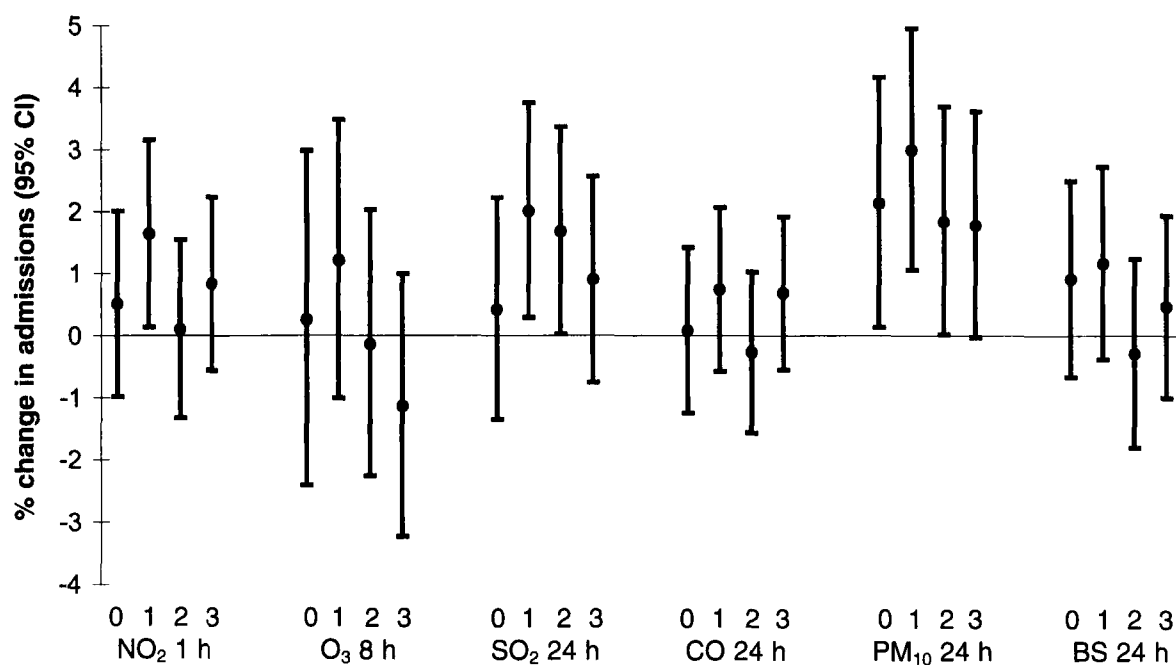


Fig. 1. Single-pollutant model results for respiratory disease (all ages). Associations are expressed as percentage change (95% confidence interval [CI]) in number of admissions for a 10th- to 90th-percentile change in each pollutant. Results for each lag tested, lag 0 to lag 3, for each pollutant are shown.

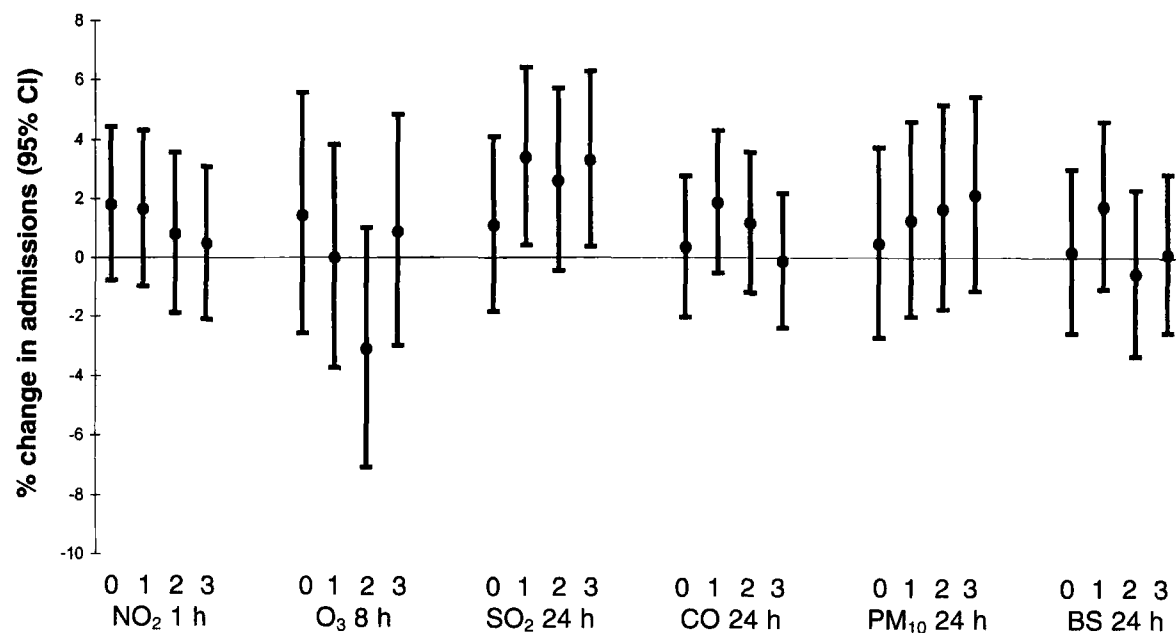


Fig. 2. Single-pollutant model results for asthma (all ages). Associations are expressed as percentage change (95% confidence interval [CI]) in number of admissions for a 10th- to 90th-percentile change in each pollutant. Results for each lag tested, lag 0 to lag 3, for each pollutant are shown.

time between measuring the pollutant and the admission increased. This pattern was reversed for O_3 .

All cardiovascular diseases. The associations between the various air pollutants and admissions for all cardiovascular diseases are summarized in Table 3. Re-

sults for the all-ages group for each single-day lag tested are illustrated in Figure 5. Several pollutants were associated positively with increases in admissions for cardiovascular disease on the same day as the pollutant was measured. These associations occurred across the

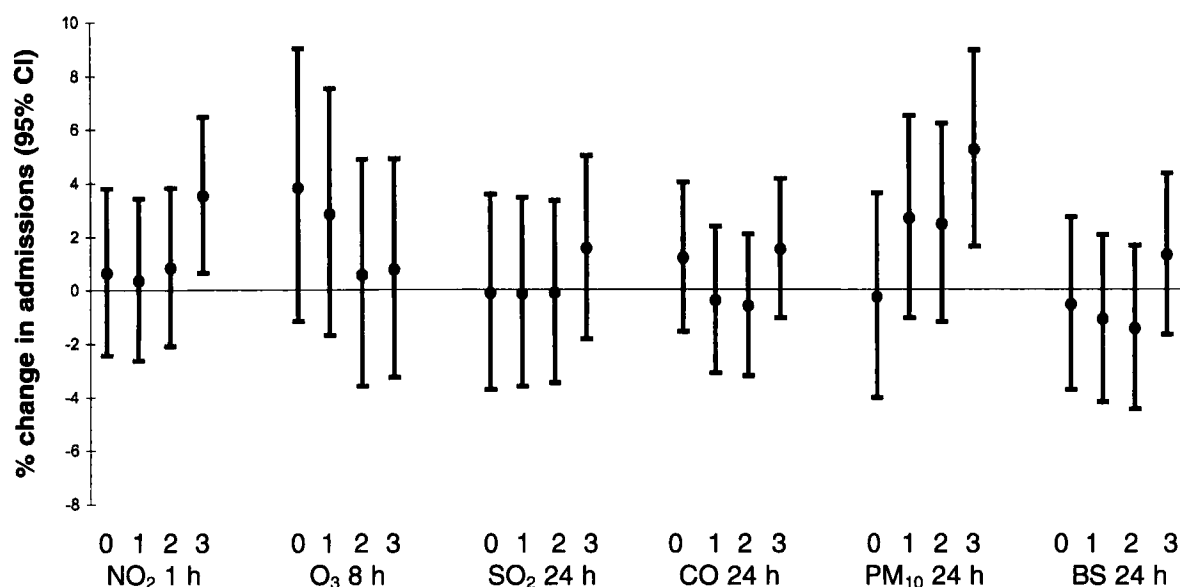


Fig. 3. Single-pollutant model results for chronic obstructive pulmonary disease (COPD) + asthma (65+ y of age). Associations are expressed as percentage change (95% confidence interval [CI]) in number of admissions for a 10th- to 90th-percentile change in each pollutant. Results for each lag tested, lag 0 to lag 3, for each pollutant are shown.

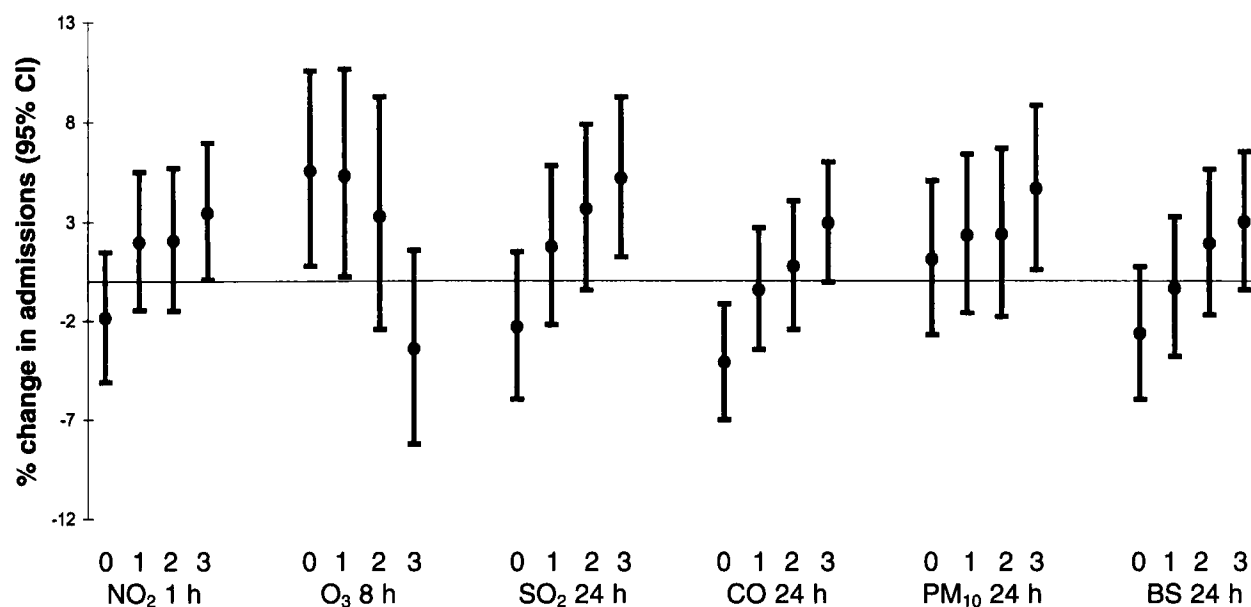


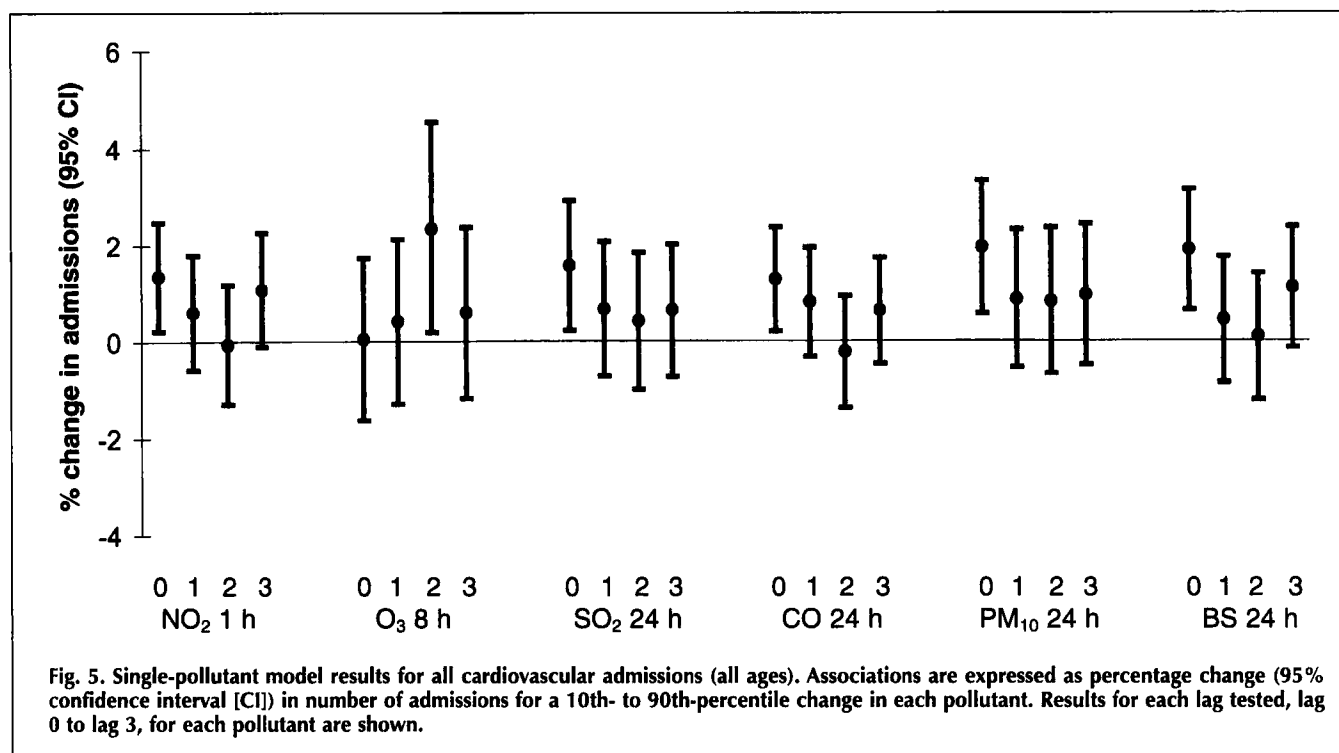
Fig. 4. Single-pollutant model results for lower respiratory infections (65+ y of age). Associations are expressed as percentage change (95% confidence interval [CI]) in number of admissions for a 10th- to 90th-percentile change in each pollutant. Results for each lag tested, lag 0 to lag 3, for each pollutant are shown.

age groups, but they were especially evident in the elderly. These associations were quite small, typically around 2% or 3%. The strongest associations were found in the elderly, where BS and O₃ in particular showed strong associations on the same day as the admission (BS 24-h, 2.71%; 95% CI = 1.22, 4.23; $p = .0004$; and O₃ 24-h, 3.38%; 95% CI = 1.12, 5.69; $p = .003$).

Ischemic heart disease. In the elderly population, all pollutants, except O₃, were associated positively with

increases in admissions for ischemic heart disease (Table 3). These associations were small in magnitude, typically around 3%, and were only marginally significant. There was little evidence of significant associations between admissions in children and adults and air pollution.

Seasonal differences. For all respiratory diseases, there were significant seasonal differences for NO₂ ($p = .01$), CO ($p = .02$), and PM₁₀ ($p = .005$); such differences did not exist for O₃ or BS. The effect estimates were sig-



nificantly larger in the warm season: NO₂ (4.63%; 95% CI = 1.87, 7.46); CO (3.79%; 95% CI = 0.89, 6.77); and PM₁₀ (6.45%; 95% CI = 3.34, 9.65). These seasonal differences were stronger in children during the warm season; the effect of SO₂ was also found to be significantly elevated ($p = .008$), compared to in the cool season (8.83%; 95% CI = 4.87, 12.94, respectively). The only other seasonal differences found were in the older elderly population (75+ y) for SO₂ and PM₁₀—both having significantly more profound effects in the cool season: (6.02% [95% CI = 2.21, 9.96] and 7.16% [95% CI = 3.15, 11.32], respectively).

For all cardiovascular diseases, there were significant seasonal differences in the elderly population for NO₂ and PM₁₀. Both pollutants showed increased effect estimates in the cool season, compared with the warm season.

Two-pollutant models. Given that both SO₂ and PM₁₀ showed strong statistically significant associations with respiratory admissions in children, and to a lesser degree in adults and the elderly, we fitted a series of two-pollutant models to determine whether the associations were independent of each other. When SO₂ and PM₁₀ were included in the same model, the magnitude of the individual associations were much reduced, as were their statistical significance. This reduction occurred in children, adults, and the elderly. The other pollutants all had the effect of reducing the magnitude of the individual SO₂ and PM₁₀ associations, although their statistical significance was unaffected. Again, this pattern occurred in each of the three main age groups. It is clear from these results that both SO₂ and PM₁₀ were indicators of the same pollutant mixture.

To determine whether the same-day associations for

all cardiovascular diseases and the pollutants reported in the 65+ age group were a reflection of some general pollutant mixture or a specific response to an individual pollutant, we ran a series of two-pollutant models. The associations with NO₂, SO₂, CO, and PM₁₀ were all moderated by the presence of BS in the model, whereas the BS association was largely unaffected by these other pollutants. In all cases, O₃ had little impact on the other pollutant effects.

Pollutant/temperature/humidity interactions. Figures 6 and 7 illustrate for respiratory and cardiovascular outcomes, respectively, the percentage change in admissions and 95% confidence intervals associated with each of the 6 pollutants investigated at each of the 3 tertiles of the temperature and humidity distributions; long-term temporal patterns in the data were controlled for. With respect to respiratory admissions, there was clearly a trend towards higher pollutant effect estimates as temperature increased. There was also a tendency for a reduction in the pollutant effect estimates on days when humidity was in the lower tertile. For cardiovascular admissions, the pattern of pollutant effects across tertiles of temperature and humidity were less systematic.

Dose-response relationships. Figure 8 illustrates, with the aid of bubble plots, the dose-response relationships for both respiratory and cardiovascular admissions and all 6 pollutants. Bubble plots summarize the relationship between 2 variables by plotting the mean of 1 variable, in this case residuals from a model, against the mean of the 2nd variable, pollutant levels, at specified intervals. The relative positions of these mean values are represented by the locations of the center of the bubbles, and the diameter of the bubbles indicates the number of observations within the intervals. Figure 8

provides evidence that the relationships between admissions and the pollutants were approximately linear, with little evidence of a threshold effect. There was some suggestion that the NO_2 associations were nonlinear.

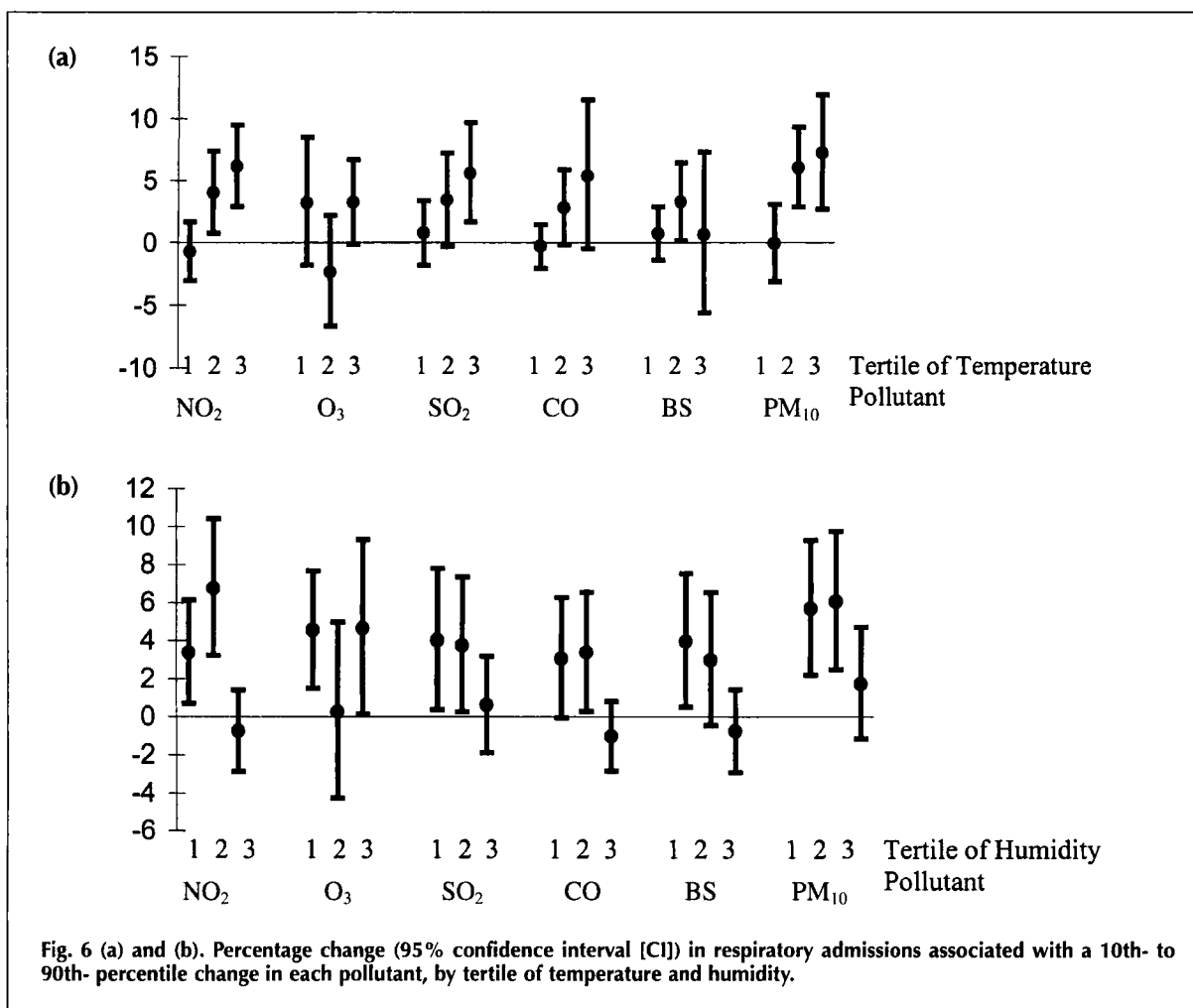
Discussion

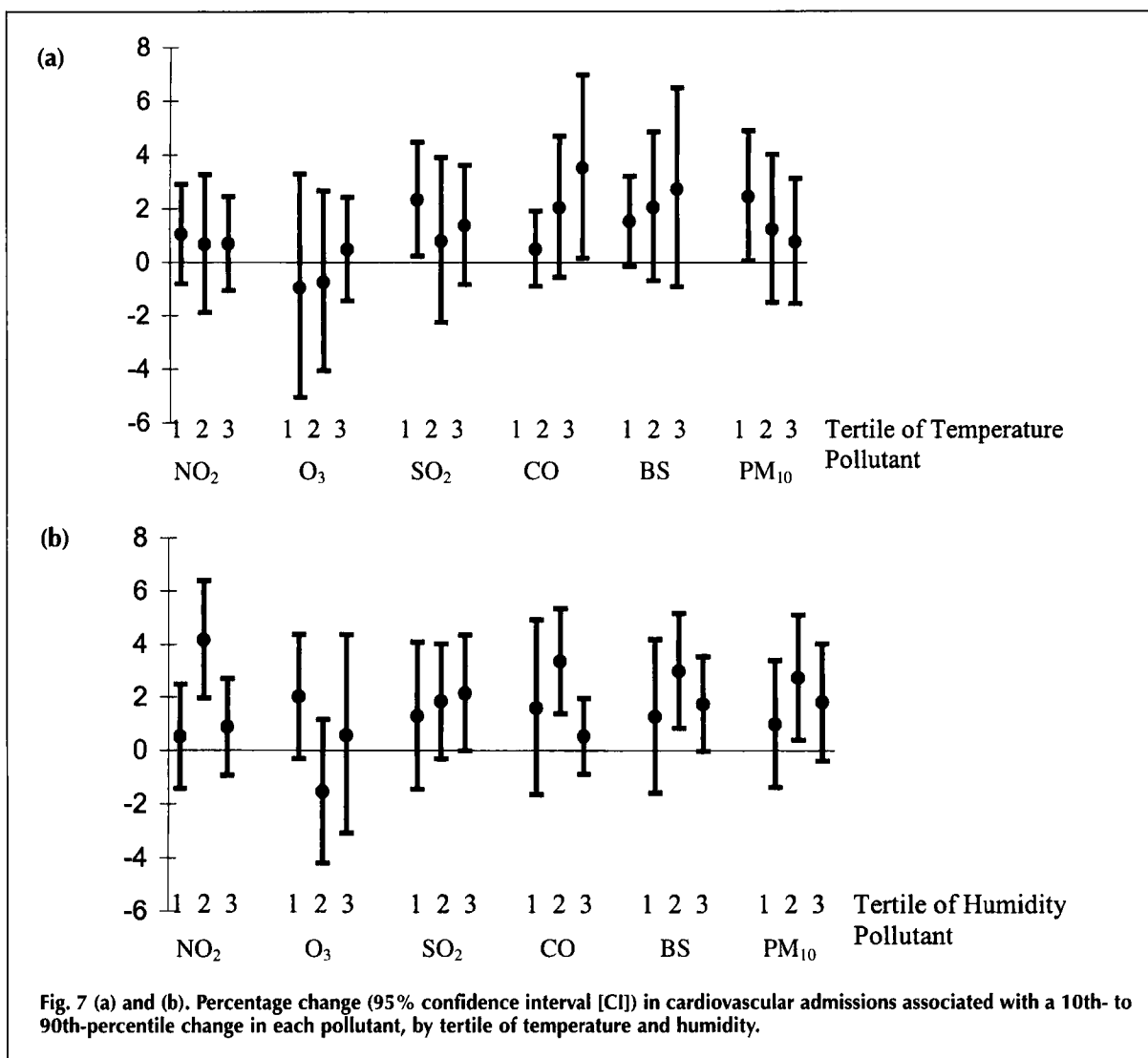
In this study, we used accepted techniques to carry out a comprehensive and methodological assessment of the associations between emergency hospital admissions and air pollution. Studying a large city like London enabled us to investigate a wide range of diagnoses and age groups while retaining statistical power. The period 1992 to 1994 was chosen for two reasons. First, PM_{10} data in London were available (from 1992), thus allowing us to make direct comparisons with results for BS and to make comparisons with the results for PM_{10} from America. Second, by investigating a later time period than that used in the earlier London study, we were able to evaluate the consistency of the associations over time.

Multiple-significance testing. Given the large number of analyses undertaken, greater emphasis has been placed on results that were highly statistically significant (i.e., $p < .01$). To avoid placing too much emphasis

on isolated significant associations, we placed more importance on statistically significant results, which were also consistent in terms of direction and magnitude across the various measures and lags of each pollutant tested.

Population exposure to air pollution. A limitation of our study was that the population exposure to outdoor air pollution across London, an area covering approximately 1 600 km^2 , was derived from only a few monitoring stations. Only "urban background" monitors were used in the study, thus avoiding monitors close to local sources as well as roadside monitors. The correlation between sites for NO_2 , O_3 , and CO were all positive and high (i.e., 0.7–0.96), suggesting that although levels may have varied between locations, the pattern of fluctuations did not. Correlations between the 5 sites providing SO_2 and BS data were smaller (i.e., 0.5–0.8 for BS and 0.2–0.5 for 4 of the 5 SO_2 sites). To what extent these monitors provide "good" estimates of the average population exposure is unknown. One additional station, located in the southeastern part of the study area, monitored data for NO_2 , CO, PM_{10} , and O_3 for only part of the period of investigation. We used its data to compare measurements with those from the centrally located monitors. The correlations exceeded





0.8 for all 4 pollutants, suggesting a uniformity of daily variation across a wide geographical area that encompassed intercity and suburban areas. However, absolute levels of NO₂, PM₁₀, and CO were lower at the periphery of London than its center, as expected; the reverse was true for O₃. These findings suggest that although levels may vary across a study area, daily variations do not; therefore, it is reasonable for one to use daily averages from a small number of monitoring stations as indicators of regional background exposure for the study population.

Comparisons with other studies. One of the main aims of our study was the testing of the hypotheses generated by an analysis of air pollution and hospital admissions in London between 1987 and 1992. In addition, we compared our findings with those from a meta-analysis of the European cities included in the APHEA project, a study in Birmingham, England, as well as with several studies from America. Relative risk estimates for all respiratory all-age admissions associated with a 100- $\mu\text{g}/\text{m}^3$ increase in particles (i.e., PM₁₀), O₃, NO₂ and SO₂ for several studies are shown in Tables 4–6. Results

for each of these pollutants are discussed in turn. We also compared our findings for ischemic heart disease and PM₁₀ with two American studies and the Birmingham study.

Associations between admissions for respiratory disease and particles (Table 4). The PM₁₀ data were not available for the London 1987–1992 and APHEA analyses; therefore, we limited direct comparisons for PM₁₀ to the American studies and a study conducted in Birmingham, England. In Schwartz et al.'s American study, they calculated a weighted average for the relative risk associated with an increase of 100 $\mu\text{g}/\text{m}^3$ in PM₁₀. As is shown in Table 4, the PM₁₀ result from the present study is in very close agreement with their results. Schwartz et al. argued that a nonsignificant test for heterogeneity between studies indicated that confounding by other pollutants or weather was not the source of these associations. When our result was included and heterogeneity retested, a nonsignificant result was obtained: $\chi^2 = 2.89$ (6 degrees of freedom [df], $p = .8$), thus confirming the consistency of our London result for PM₁₀ with those from America reported by Schwartz et al. The Bir-

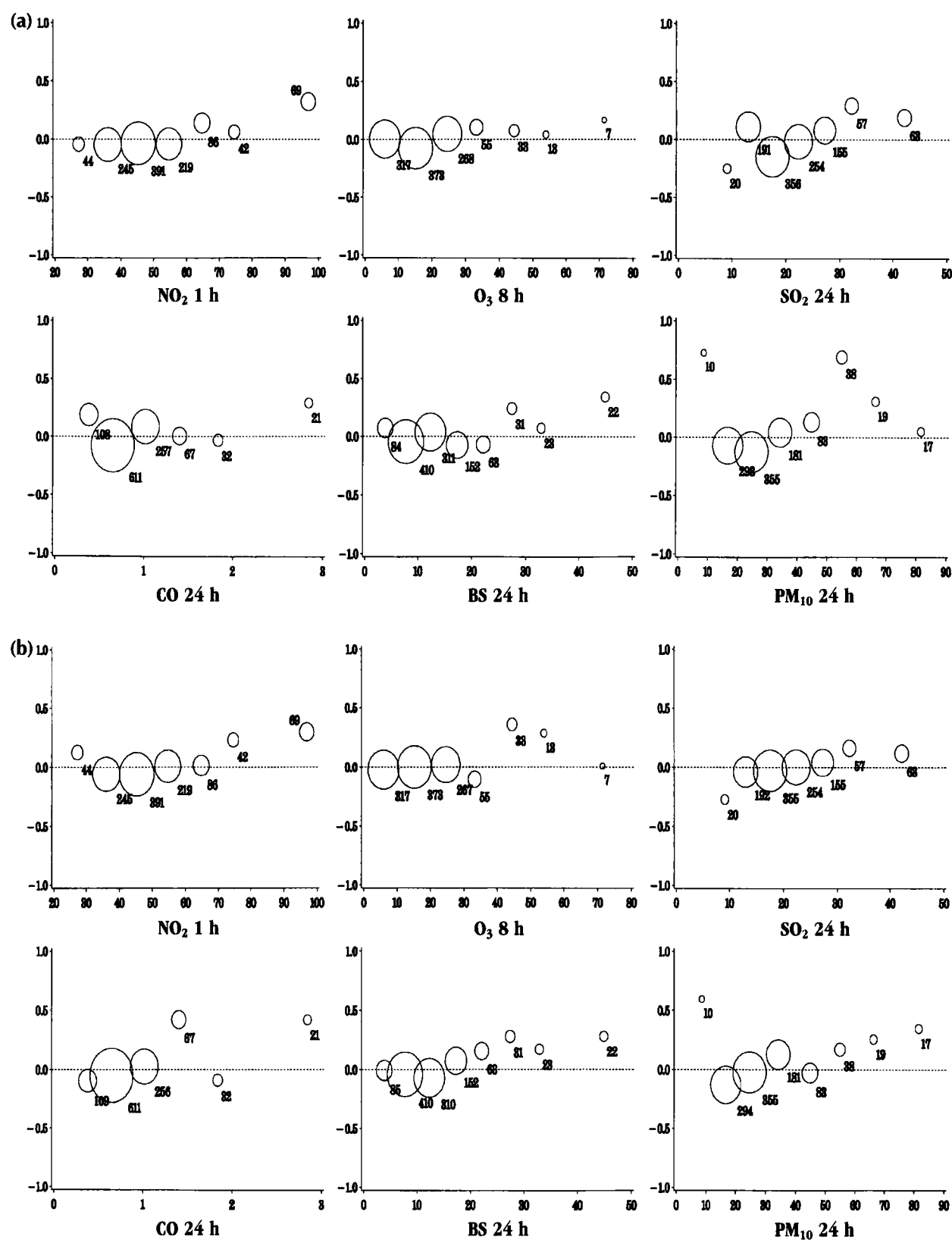


Fig. 8 (a) and (b). Bubble plots illustrating dose-response relationships for all respiratory and all cardiovascular admissions. The mean values for model residuals (including seasonal patterns, temperature, and humidity—but excluding air pollution) are plotted against mean pollutant values for given pollutant intervals. The relative positions of these mean values are represented by the locations of the center of the bubbles, and the diameter of the bubbles indicates the number of observations within each interval.

mingham estimate for PM_{10} was much higher than the estimate for London, and the summary American estimate and its confidence interval barely overlap the American figure.

Our result for BS was consistent with those found in the London 1987–1992 and APHEA meta-analyses (which include the earlier London results) (Table 4). The effect estimates are comparable in magnitude and direction and are statistically nonsignificant. For studies of all respiratory diseases, the associations with PM_{10} were generally larger and more statistically significant than the associations with BS. These results suggest that there is some component(s) of the PM_{10} particle fraction that goes unmeasured by the BS reflectance techniques; perhaps this component has an adverse effect on the respiratory system.

Associations between admissions for respiratory disease and O_3 . Our results for O_3 (Table 5) were substantially different from those reported in the earlier London study, APHEA project, and America. We reported a reduction, albeit nonsignificant, in admissions for respiratory disease associated with an increase in O_3 . However, the data in Table 5 are a little misleading as there are small methodological differences between the current London study and the earlier London and APHEA project studies. The reported result for the present study was selected because it was the most significant association (irrespective of direction) from the various lags of 1-h O_3 tested (lag 0 to lag 3). The quoted result for the London 1987–1992 analysis was for the 8-h measure for O_3 lagged 1 d. It was selected as the most significant positive association from the lags investigated (lags 0, 1, and 2 d). In our study, the most significant positive association from lags 0 to 2 for the 8-h O_3 (and 1-h O_3) are also the lag 1-d measures. The relative risk associated with a $100\text{-}\mu\text{g}/\text{m}^3$ increase in 1-h O_3 , lag 1 d, in our study was 1.07 (95% CI = 0.93, 1.24)—a nonsignificant result that is not inconsistent with the earlier London finding of 1.39 (95% CI = 1.14, 1.71) and similar to the American summary figure of 1.06 (95% CI = 1.05, 1.08). Including this O_3 estimate with the American estimates and reassessing the heterogeneity produced a nonsignificant result ($\chi^2 = 9.17$, 6 df, $p = .2$), which suggested consistency between our present result for O_3 and those reported by the Americans.

Associations between admissions for respiratory disease and SO_2 and NO_2 . The results for SO_2 (Table 6) from London for 1987–1992 and the APHEA project (which included the earlier London result) show smaller effect estimates than those obtained from the present study. The earlier London estimate is within the 95% confidence interval for our result, so the two are not significantly different. In children, the association with SO_2 was particularly strong; this was not the case, however, in the 1987–1992 London study. The results from Birmingham were not available, but they are reportedly nonsignificant. The results for NO_2 from the two London studies were similar and tended to give larger effect estimates than the combined APHEA project estimates.

Associations between admissions for COPD plus asthma and particles/ O_3 . For COPD admissions,

Schwartz et al. calculated a weighted mean relative risk of 1.19 (95% CI = 1.15, 1.34) for a $100\text{-}\mu\text{g}/\text{m}^3$ increase in PM_{10} . This estimate was based upon 7 studies from America and Spain. The APHEA project estimate (again for all ages) was based upon BS; the relative risk for a $100\text{-}\mu\text{g}/\text{m}^3$ increase was 1.04 (95% CI = 1.01, 1.06). In our study, we examined COPD admissions and asthma admissions collectively in only the elderly. Our estimates for the same increase in BS and PM_{10} were relative risks of 0.91 (95% CI = 0.75, 1.11) and 1.18 (95% CI = 1.05, 1.32), respectively. Whereas our estimate for BS was in the opposite direction of that reported by the APHEA project meta-analysis (despite the fact that confidence intervals overlapped), our result for PM_{10} showed a degree of consistency with the American findings. Our estimate for the effect of O_3 was larger (and nonsignificant) than both the American and APHEA project estimates, but it had a wide confidence interval

Table 4.—Associations between All Age Groups, All Respiratory Admissions, and Particles

Location and study	Particle measure	RR*	95% CI
Buffalo (Thurston et al., 1992)†	PM_{10}	1.24	1.06, 1.45
New York (Thurston et al., 1992)	PM_{10}	1.10	1.02, 1.20
Ontario (Burnett et al., 1994)‡	PM_{10}	1.12	1.08, 1.15
New Haven (Schwartz, 1995b)	PM_{10}	1.12	1.01, 1.26
Spokane (Schwartz, 1995a)	PM_{10}	1.17	1.07, 1.27
Tacoma (Schwartz, 1995b)	PM_{10}	1.20	1.06, 1.35
Weighted average§	PM_{10}	1.13	1.10, 1.15
London (present study)¶	PM_{10}	1.10	1.03, 1.17
London (present study)¶	BS	1.07	0.98, 1.18
London (Ponce de Leon et al., 1996)//	BS	1.02	0.92, 1.14
Birmingham, England (Walters et al., 1994)**	PM_{10}	1.26	1.12, 1.44
APHEA (Spix et al., 1998)††			
65+ y of age	BS	1.04	0.99, 1.09
APHEA (Spix et al., 1998)††			
15–65 y of age	BS	1.06	1.01, 1.10

Notes: CI = confidence interval, PM_{10} = particulate matter 10 μm or less in diameter, RR = relative risk, BS = Black Smoke, and APHEA = Air Pollution and Health: a European Approach project.

*RR associated with a $100\text{-}\mu\text{g}/\text{m}^3$ increase in 24-hr average measure of particles.

†Based on Six-City Study data; converted, assuming $SO_4/PM_{10} = 0.25$.

‡Converted, assuming $SO_4/PM_{10} = 0.25$.

§Selected studies from America and weighted average.²⁵ Most of the American studies quoted were based on two-pollutant models, including ozone (O_3) and PM_{10} .

¶Single-pollutant all-ages result. In two-pollutant models for age subgroups, O_3 does not affect the single-pollutant model results significantly.

//Black Smoke, in presence of O_3 for 8 hr; all ages.

**Single-pollutant result; all ages.

††Black Smoke, single-pollutant result, meta-analysis for four European cities: London, Amsterdam, Rotterdam, and Paris.

Table 5.—Associations between All Ages, All Respiratory Admissions, and Ozone

Location and study	RR*	95% CI
Buffalo (Thurston et al., 1992)	1.12	0.99, 1.24
New York (Thurston et al., 1992)	1.07	1.04, 1.09
Ontario (Burnett et al., 1994)	1.05	1.03, 1.07
New Haven (Schwartz, 1995b)†	1.07	0.99, 1.15
Spokane (Schwartz, 1995a)	1.54	1.00, 2.38
Tacoma (Schwartz, 1995b)†	1.20	1.06, 1.37
Weighted average‡	1.06	1.05, 1.08
London (present study)§	0.90	0.78, 1.03
London (Ponce de Leon et al., 1996)#	1.39	1.14, 1.71
Birmingham, England (Walters et al., 1994)//	NS	
APHEA (Spix et al., 1998)** 65+ y	1.06	1.03, 1.10
APHEA (Spix et al., 1998)** 15–65 y	1.04	1.01, 1.07

Notes: RR = relative risk, CI = confidence interval, NS = not significant, and APHEA = Air Pollution and Health: a European Approach project.

*RR associated with a 100- $\mu\text{g}/\text{m}^3$ increase in peak hour ozone (O_3).

†Peak/mean O_3 of 2.5.

‡Selected studies from America and weighted average.²⁵ Most of the American studies quoted were based on two-pollutant models, including O_3 and particulate matter 10 μm or less in diameter (PM_{10}).

§Most-significant single-pollutant result; all ages. Converted ppb to $\mu\text{g}/\text{m}^3$ by 2 $\times$.

#Most-significant single-pollutant result; all ages. Converted ppb to $\mu\text{g}/\text{m}^3$ by 2 $\times$ and mean/8 hr 1.5 $\times$.

//No figures available, but reported as nonsignificant.

**Single-pollutant results.

Table 6.—Associations between All Ages, All Respiratory Admissions, and Nitrogen Dioxide (NO_2) and Sulfur Dioxide (SO_2)

Location and study	NO_2 (1 hr)		SO_2 (24 hr)	
	RR	95% CI	RR	95% CI
London (present study)*	1.09	1.01, 1.18	1.12	1.02, 1.23
London (Ponce de Leon et al., 1996)†	1.13	1.00, 1.27	1.03	0.94, 1.14
Birmingham, England (Wordley et al., 1997)‡	NS		NS	
APHEA (Spix et al., 1998) 65+ y	1.01	0.95, 1.07	1.04	1.01, 1.09
APHEA (Spix et al., 1998) 15–65 y	1.01	0.99, 1.02	1.02	0.98, 1.05

Notes: RR = relative risk, CI = confidence interval, NS = not significant, and APHEA = Air Pollution and Health: a European Approach project.

*Conversion factor of 1.913 ppb to $\mu\text{g}/\text{m}^3$ for NO_2 .

†Conversion factor of 1.54 for 1 hr to 24 hr for NO_2 .

‡No figures available, but reported as nonsignificant.

overlapping with both of the confidence intervals from these studies. With respect to asthma alone, our results were consistent with those of a meta-analysis of the APHEA project cities.¹⁰

Associations between admissions for ischemic heart disease and particles. Schwartz et al. quoted 2 results for PM_{10} from Ontario and Detroit for ischemic heart disease admissions; (1) a relative risk for a 100- $\mu\text{g}/\text{m}^3$ increase of 1.07 (95% CI = 1.02, 1.12) and a relative risk of 1.05 (95% CI = 1.02, 1.10), respectively. These are consistent with our result for PM_{10} : 1.10 (95% CI = 1.02, 1.20). The Birmingham group did not find any significant associations in their study between PM_{10} and hospital admissions.

Interrelationships between temperature, humidity, and pollutants. An important factor in assessing the potential effect of air pollution on hospital admission rates is the interrelationship between daily measures of air pollution and meteorological factors (e.g., temperature, humidity). How these environmental factors interact may determine their individual and composite effect on admissions. Goldsmith et al.,²⁶ who used conventional correlation and regression techniques, were unable to distinguish the effects of pollution by particulate sulphate from those resulting from meteorological variables and oxidants. Samet et al.²⁷ compared approaches for controlling for effects of weather variables when they analyzed the connection between air pollution and daily mortality. They could find no meaningful pattern of variation in the relative risks calculated under different weather conditions, regardless of the approach they used to characterize weather. One such method was the synoptic approach proposed by Kalkstein et al.,²⁸ for which numerous weather variables are combined to provide specific characterizations of weather conditions.

In our study, we were primarily interested in whether there were effects of air pollution independent of meteorological factors. As a result, our approach to this problem was to first investigate the relationships between admissions and temperature and humidity. Once these influences had been modeled successfully, there remained negligible associations between admissions (model residuals) and temperature and humidity. Only then were air-pollution variables introduced into the statistical model. We acknowledge that in using this approach, we may have neglected some of the details of how these environmental factors combine and interact and we may have missed an effect of air pollution because it was perhaps present only under certain meteorological conditions. Conversely, our approach ensured that we did not misinterpret the effects of temperature and humidity on admissions as effects of air pollution.

In Figure 6, one notes that there was clearly a trend toward higher pollutant-effect estimates as temperature increased. It is unclear whether this is evidence of a significant interaction between air pollution and temperature. The results of the seasonal analysis (see Results) indicate that effect estimates appeared stronger in the warmer months (April to September) than the remaining cooler months—even when temperature and humidity were allowed for. Either this seasonal difference result-

ed from inappropriate control for meteorological conditions or there were other factors, such as time spent outside, which had an impact on the magnitude of the pollutant effects.

Conclusions

Increases in the ambient levels of each of the pollutants studied were associated with short-term increases in hospital admissions for one or more disease groups. Whereas PM₁₀ and SO₂ were most evidently associated with respiratory diseases in children, CO and BS were associated with cardiovascular disease—although weaker associations existed with the other pollutants. Of the two measures of particulate pollution, PM₁₀ appeared to have stronger associations with increases in emergency admissions than BS. The associations between respiratory diseases and PM₁₀ and O₃ were comparable to those reported in North American studies and in those from Europe. They were also consistent with a previous analysis of London for a different time period.

Although there was some suggestion that the "effects" of air pollution on respiratory admissions increased with temperature, we were unable to determine whether this was, in fact, an interaction with temperature or an interaction with some other seasonal factor. For both respiratory and cardiovascular admissions, we found little evidence for the existence of thresholds of response to pollutants. The one possible exception was respiratory admissions and NO₂ for which a nonlinear relationship was found.

This study is part of an ongoing project, funded by the Department of Health (JR 121/6267), to investigate the effects and likely health care costs of outdoor air pollution on mortality, hospital admissions, visits to A&E departments, and GP consultations in London. The project team comprises Professors H.R. Anderson (Principal Investigator); D.P. Strachan; J.M. Bland; Mr. R.W. Atkinson and Mr. S. Bremner from St George's Hospital Medical School, London; Professor A. Haines and Mr. S. Hajat from University College London, London; Professor A. McMichael from London School of Hygiene and Tropical Medicine, London; Dr. J.S. Bower from AEA NETCEN, Oxfordshire; and Dr. J. Emberlin from the Pollen Research Unit, Worcester. We are grateful for the statistical advice and assistance and computing support from Roger Newson, Jan Poloniecki, Liz Limb, and Iain Carey.

Submitted for publication June 18, 1998; revised; accepted for publication December 7, 1998.

Requests for reprints should be sent to R. W. Atkinson, Department of Public Health Sciences, St. George's Hospital Medical School, Cranmer Terrace, London SW17 0RE.

References

1. Ministry of Health. Mortality and morbidity during the London fog of December 1952. Reports on Public Health and Medical Subjects No 95. London: HMSO, 1954.
2. Dockery DW, Pope CA. Acute respiratory effects of particulate air pollution. *Ann Rev Public Health* 1994; 15:107–32.
3. Schwartz J. Air pollution and hospital admissions for the elderly in Detroit, Michigan. *Am J Respir Crit Care Med* 1994; 150:648–55.
4. Schwartz J. Air pollution and hospital admissions for the elderly in Birmingham, Alabama. *Am J Epidemiol* 1994; 139:589–98.
5. Schwartz J, Morris R, Moolgavkar SH, et al. Air pollution and hospital admissions for cardiovascular disease in Detroit, Michigan. A critical review of the evidence on particulate air pollution and mortality. *Am J Epidemiol* 1995; 142:23–35.
6. Moolgavkar SH, Luebeck EG. A critical review of the evidence on particulate air pollution and mortality. *Epidemiology* 1996; 7:420–28.
7. Lebowitz MD. Epidemiological studies of the respiratory effects of air pollution (Review). *Eur Respir J* 1996; 9:1029–54.
8. Anderson HR, Spix C, Medina S, et al. Air pollution and daily admissions for chronic obstructive pulmonary disease in 6 European cities: results from the APHEA project. *Eur Respir J* 1997; 10:1064–71.
9. Touloumi G, Katsouyanni K, Zmirou D, et al. Short-term effects of ambient oxidant exposure on mortality: a combined analysis within the APHEA project. *Air Pollution and Health: a European Approach*. *Am J Epidemiol* 1997; 146:177–85.
10. Sunyer J, Spix C, Quenel P, et al. Urban air pollution and emergency admissions for asthma in four European cities: the APHEA Project. *Thorax* 1997; 52:760–65.
11. Spix C, Anderson HR, Schwartz J, et al. Short-term effects of air pollution on hospital admissions for respiratory diseases in Europe: a quantitative summary of APHEA study results. *Arch Environ Health* 1998; 53:54–64.
12. Schwartz J. Air pollution and daily mortality: a review and meta-analysis. *Environ Res* 1994; 64:36–52.
13. Dockery DW, Pope CA. Acute respiratory effects of particulate air pollution. *Ann Rev Public Health* 1994; 15:107–32.
14. American Thoracic Society. Health effects of outdoor air pollution. *Am J Respir Crit Care Med* 1996; 153:3–50.
15. Brunekreef B, Dockery DW, Krzyzanowski M. Epidemiologic studies on short-term effects of low levels of major ambient air pollution components. *Environ Health Perspect* 1995; 103:3–13.
16. Katsouyanni K, Schwartz J, Spix C, et al. Short-term effects of air pollution on health: a European approach using epidemiologic time series data: the APHEA protocol. *J Epidemiol Commun Health* 1996; 50(Suppl 1):S12–S18.
17. Katsouyanni K, Zmirou D, Spix C, et al. Short-term effects of air pollution on health: a European approach using epidemiological time-series data. The APHEA project: background, objectives, design. *Eur Respir J* 1995; 8:1030–38.
18. Ponce de L, Anderson HR, Bland JM, et al. Effects of air pollution on daily hospital admissions for respiratory disease in London between 1987–88 and 1991–92. *J Epidemiol Commun Health* 1996; 50(Suppl 1):s63–s70.
19. Wordley J, Walters S, Ayres JG. Short-term variations in hospital admissions and mortality and particulate air pollution. *Occup Environ Med* 1997; 54:108–16.
20. Buck SF. A method of estimation of missing values in multivariate data suitable for use with an electronic computer. *R.S.S.(B)* 1960; 22:302–06.
21. AEA Technology: National Environmental Technology Center. Air Pollution in the UK: 1994. AEA/RAMP/200015001/1 1997.
22. Durrleman S, Simon R. Flexible regression models with cubic splines. *Stat Med* 1989; 8:551–61.
23. Zeger SL. A regression model for time series of counts. *Biometrika* 1988; 75:621–29.
24. SAS. Cary, NC: SAS Institute, 1997.
25. Schwartz J, Fletcher T, McMichael AJ (Eds). Health at the Crossroads: Transport Policy and Urban Health. Health Effects of Air Pollution from Traffic: Ozone and Particulate Matter; 1997.
26. Goldsmith JR, Griffith HL, Detels R, et al. Emergency room admissions, meteorologic variables, and air pollutants: a path analysis. *Am J Epidemiol* 1983; 118:759–78.
27. Health Effects Institute. Particulate air pollution and daily mortality. Analyses of the effects of weather and multiple air pollutants. Phase I.B. Report of the Particle Epidemiology Evaluation Project. Health Effects Institute, 1997.
28. Kalkstein LS, Tan G, Skindlov J. An evaluation of objective clustering procedures for use in synoptic climatological classification. *J Climate Appl Meteorol* 26:717–30.